

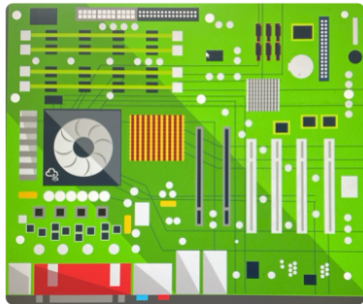
Operating System Security :— (This room introduces users to operating system security and demonstrates SSH authentication on linux).

Task 1 :— Introduction to operating system security :—

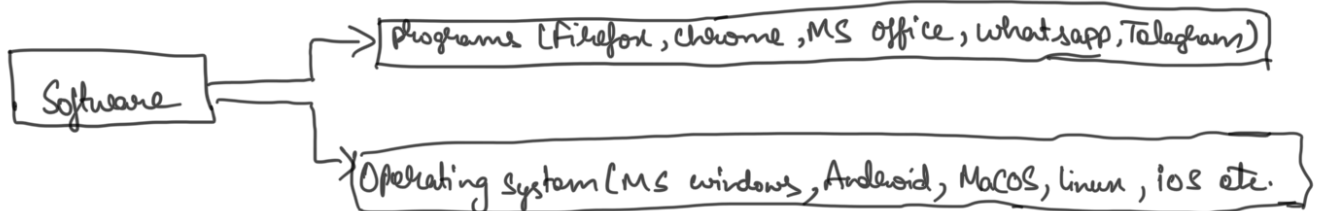
Every day you use a smartphone or laptop or almost any type of computer, you interact directly or indirectly with an operating system. Operating systems include MS Windows, macOS, iOS, Android, Chrome OS and Linux. But what is OS? To define an OS, we need to visit computer terms: ^{hardware} Computer hardware refers to all computer parts and peripherals that you can touch with your hand.

Hardware includes Screen, Keyboard, Printer and USB flash memory, and desktop board.

As shown below, desktop board contains many components, in particular, A central processing unit (CPU) and Memory chip (RAM). The desktop board is usually connected to a storage device (HDD or SSD).



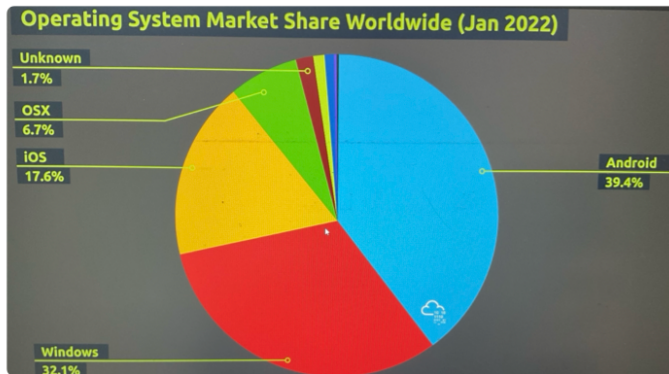
The desktop board is main part of the computer, and all other pieces of hardware from keyboard and mouse to screen and printer connect to it. However, hardware components by themselves are useless if you want to run your favourite program and applications. We need OS to control & drive them.



The OS is layer sitting b/w hardware and applications and programs you are running. Example programs you use daily might include web browser, such as Firefox, Safari and Chrome. All programs and applications cannot run directly on computer hardware, however they run on top of operating system. The operating system allows these programs to access hardware according to specific rules. Some OS are designed to run on laptops and personal desktops, such as Windows 11 and macOS. Other OS are designed specifically for smartphones, such as Android and iOS. There is also OS intended for servers. Examples include MS Windows Server 2019, IBM AIX and Oracle

Finally there are OS that they can use on personal computer and server, one example is linux.

The image below shows popularity of different OS used to browse Internet on data collected during 2022.



Your smartphone might be running on Android or iOS, you might have plenty of private data on it. Example:

1. private conversations with your family and friends
2. private photos with family and friend
3. Email client that you use for personal and work communications.
4. passwords saved in web browser (or even in notes).
5. E-banking apps

The list of confidential and private data goes on. You don't want someone you don't trust to open your phone and go through your photos, conversations and apps. Hence you need to secure your phone and its operating system.

The same goes for your laptop or computer running MS Windows, macOS or Linux. Your computer most likely contains plenty of information.

1. Confidential files related to your work or university
2. private personal files, such as copy of your ID or passport.
3. Email programs, such as MS Outlook and Mozilla Thunderbird
4. passwords saved in web browser or other apps.
5. Copy of digital camera and smartphone photos.

You want to ensure that your data is secure. When we talk about security, we should think of protecting 3 things.

Confidentiality: — You want to ensure that secret and private files and information are only available to intended persons. (Specific individual for whom something is meant or directed).

Integrity: — It is crucial that no one can tamper with files stored on your system or while being transferred on network.

Availability: — You want your laptop or smartphone to be available to use anytime you decide to use it.



Quiz :-

Which of following is not operating system?

Ⓐ AIX Ⓐ Chrome OS

Ⓐ Android Ⓐ Solaris Ⓐ Thunderbird

A) Thunderbird